

3.8 Implicit Differentiation Worksheet

1. Find the slope of the line tangent to the graph of the ellipse $\frac{x^2}{2} + \frac{y^2}{36} = 1$ at the point $\left(\frac{3}{2}, 3\right)$.

We calculate the derivative: $\frac{d}{dx}\left[\frac{x^2}{2} + \frac{y^2}{36}\right] = \frac{d}{dx}[1] \rightarrow x + \frac{y}{18} \cdot \frac{dy}{dx} = 0 \rightarrow \frac{dy}{dx} = -\frac{18x}{y}$

Therefore the slope of the tangent line is: $m = -\frac{18\left(\frac{3}{2}\right)}{(3)} = -\frac{27}{3} = -9$.

2. Find the equation of the line tangent to the curve $5y^4 - x^3 = x^2y - 7$ at the point $(2, 1)$.

We calculate the derivative: $\frac{d}{dx}[5y^4 - x^3] = \frac{d}{dx}[x^2y - 7] \rightarrow 20y^3 \frac{dy}{dx} - 3x^2 = x^2 \frac{dy}{dx} + 2xy$

$$\rightarrow 20y^3 \frac{dy}{dx} - x^2 \frac{dy}{dx} = 2xy + 3x^2 \rightarrow \frac{dy}{dx}[20y^3 - x^2] = 2xy + 3x^2 \rightarrow \frac{dy}{dx} = \frac{2xy + 3x^2}{20y^3 - x^2}$$

Therefore, the slope of the tangent line is: $\frac{dy}{dx} = \frac{2(2)(1) + 3(2)^2}{20(1)^3 - (2)^2} = \frac{4 + 12}{20 - 4} = \frac{16}{16} = 1$

Therefore, the equation of the tangent line is:

$$y - 1 = 1(x - 2)$$

$$y = x - 1$$

3. Compute $\frac{dy}{dx}$ for each of the following:

a) $xy^3 + x^5y = 42$

$$\frac{d}{dx}[xy^3 + x^5y] = \frac{d}{dx}[42] \rightarrow y^3 + x3y^2 \frac{dy}{dx} + 5x^4y + x^5 \frac{dy}{dx} = 0$$

$$\rightarrow \frac{dy}{dx}(3xy^2 + x^5) = -y^3 - 5x^4y$$

$$\rightarrow \frac{dy}{dx} = \frac{-y^3 - 5x^4y}{3xy^2 + x^5}$$

b) $\sin(x^2y^3) = 3x + 2y$

$$\cos(x^2y^3) \left[2xy^3 + x^23y^2 \frac{dy}{dx} \right] = 3 + 2 \frac{dy}{dx} \rightarrow 2xy^3 \cos(x^2y^3) + x^23y^2 \cos(x^2y^3) \frac{dy}{dx} = 3 + 2 \frac{dy}{dx}$$

$$\rightarrow x^23y^2 \cos(x^2y^3) \frac{dy}{dx} - 2 \frac{dy}{dx} = 3 - 2xy^3 \cos(x^2y^3)$$

$$\rightarrow \frac{dy}{dx} [x^23y^2 \cos(x^2y^3) - 2] = 3 - 2xy^3 \cos(x^2y^3)$$

$$\rightarrow \frac{dy}{dx} = \frac{3 - 2xy^3 \cos(x^2y^3)}{x^23y^2 \cos(x^2y^3) - 2}$$

4. Find the slope of the line tangent to the curve $x^3 + y^3 = xy^2 + 5$ at the point $(1, 2)$.

$$\frac{d}{dx}[x^3 + y^3] = \frac{d}{dx}[xy^2 + 5] \rightarrow 3x^2 + 3y^2 \frac{dy}{dx} = y^2 + x2y \frac{dy}{dx}$$

We can evaluate at the point $(1, 2)$ and then solve for $\frac{dy}{dx}$.

$$3(1)^2 + 3(2)^2 \frac{dy}{dx} = (2)^2 + (1)2(2) \frac{dy}{dx} \rightarrow 3 + 12 \frac{dy}{dx} = 4 + 4 \frac{dy}{dx} \rightarrow 8 \frac{dy}{dx} = 1 \rightarrow \frac{dy}{dx} = \frac{1}{8}$$

Therefore, the slope of the tangent line at $(1, 2)$ is $\frac{1}{8}$.

5. Compute $\frac{dw}{dq}$ given that $w^2 q^3 = \sin(w^2) + q^5$.

$$\begin{aligned} \frac{d}{dq}[w^2 q^3] &= \frac{d}{dq}[\sin(w^2) + q^5] \rightarrow 2w \frac{dw}{dq} q^3 + w^2 3q^2 = \cos(w^2) 2w \frac{dw}{dq} + 5q^4 \\ &\rightarrow 2w \frac{dw}{dq} q^3 - 2w \cos(w^2) \frac{dw}{dq} = 5q^4 - 3w^2 q^2 \\ &\rightarrow \frac{dw}{dq} [2wq^3 - 2w \cos(w^2)] = 5q^4 - 3w^2 q^2 \\ &\rightarrow \frac{dw}{dq} = \frac{5q^4 - 3w^2 q^2}{2wq^3 - 2w \cos(w^2)} \end{aligned}$$

6. If $xy = 1$ then find $\frac{dy}{dx}$.

$$\begin{aligned}\frac{d}{dx}[xy] &= \frac{d}{dx}[1] \rightarrow y + x \frac{dy}{dx} = 0 \\ &\rightarrow x \frac{dy}{dx} = -y \\ &\rightarrow \frac{dy}{dx} = -\frac{y}{x}\end{aligned}$$

7. Find $\frac{dy}{dx}$ if $5y^2 + \sin(y) = x^2$.

$$\begin{aligned}\frac{d}{dx}[5y^2 + \sin(y)] &= \frac{d}{dx}[x^2] \rightarrow 10y \frac{dy}{dx} + \cos(y) \frac{dy}{dx} = 2x \\ &\rightarrow \frac{dy}{dx}[10y + \cos(y)] = 2x \\ &\rightarrow \frac{dy}{dx} = \frac{2x}{10y + \cos(y)}\end{aligned}$$

8. Find y'' if $4x^2 - 2y^2 = 9$.

$$\begin{aligned}\frac{d}{dx}[4x^2 - 2y^2] &= \frac{d}{dx}[9] \rightarrow 8x - 4y \frac{dy}{dx} = 0 \\ &\rightarrow \frac{dy}{dx} = \frac{-8x}{-4y} = \frac{2x}{y}\end{aligned}$$

So we can say that $y' = \frac{2x}{y}$. Therefore, we can say that $y'' = \frac{d}{dx}\left(\frac{2x}{y}\right) = \frac{y(2) - \frac{dy}{dx}(2x)}{y^2}$.

We know the value of $\frac{dy}{dx}$ so we substitute it in:

$$y'' = \frac{y(2) - \left(\frac{2x}{y}\right)(2x)}{y^2} = \frac{2y - \frac{4x^2}{y}}{y^2} = \frac{2y^2 - 4x^2}{y^3} = \frac{-9}{y^3}$$

9. Consider the curve given by $x^3 + y^3 = 6xy$.

a) Find $\frac{dy}{dx}$.

$$\begin{aligned}
 \frac{d}{dx}[x^3 + y^3] &= \frac{d}{dx}[6xy] \rightarrow 3x^2 + 3y^2 \frac{dy}{dx} = 6y + 6x \frac{dy}{dx} \\
 &\rightarrow 3x^2 - 6y = 6x \frac{dy}{dx} - 3y^2 \frac{dy}{dx} \\
 &\rightarrow 3x^2 - 6y = \frac{dy}{dx}[6x - 3y^2] \\
 &\rightarrow \frac{dy}{dx} = \frac{3x^2 - 6y}{6x - 3y^2} \\
 &\rightarrow \frac{dy}{dx} = \frac{x^2 - 2y}{2x - y^2} \quad \text{or} \quad \frac{2y - x^2}{y^2 - 2x}
 \end{aligned}$$

Obtain by factoring out a negative from top & bottom

b) At what point in the first quadrant is the tangent line horizontal?

The tangent line will be horizontal when $\frac{x^2 - 2y}{2x - y^2} = 0 \rightarrow x^2 - 2y = 0 \rightarrow x^2 = 2y \rightarrow y = \frac{x^2}{2}$

We substitute this value back into the original equation.

$$\begin{aligned}
 x^3 + \left(\frac{x^2}{2}\right)^3 &= 6x\left(\frac{x^2}{2}\right) \rightarrow x^3 + \frac{x^6}{8} = 3x^3 \rightarrow \frac{x^6}{8} - 2x^3 = 0 \rightarrow x^6 - 16x^3 = 0 \\
 &\rightarrow x^3(x^3 - 16) = 0 \\
 &\rightarrow x = 0 \quad \text{or} \quad x = \sqrt[3]{16}
 \end{aligned}$$

Since $x \neq 0$ in the first quadrant we must use the value $x = \sqrt[3]{16}$. So $y = \frac{1}{2}(16)^{2/3}$.

Therefore, the tangent line is horizontal at the point $\left(\sqrt[3]{16}, \frac{1}{2}(16)^{2/3}\right)$.